BRITAIN OVER SKY

An integrated regional aviation network

A technical proposal for a national clockface timetable, British regional aircraft and connected air routes across Great Britain, Belfast, Dublin and the Isle of Man.

NETWORK DESIGN / TIMETABLE / AIRCRAFT / ECONOMICS / DELIVERY

Prepared as a strategic planning concept. It is not a published airline timetable, a detailed business case, or a claim that a new BAE Systems regional aircraft currently exists.

Executive summary

Britain Over Sky proposes treating regional air services as one coordinated transport layer. Rather than building a national map from isolated routes, the system would use a common timetable, predictable transfer windows, shared passenger information and a small number of regional interchange airports. The objective is faster cross-country travel and stronger links between the UK nations and the Republic of Ireland - not to replace rail on every journey.

Core recommendation
Start with a tightly scoped 8-corridor pilot using certified aircraft already available to the market; coordinate departure waves at Manchester, Birmingham and one Scottish hub; expand only after route-level load factors and reliability are proven.

What the system would do

- Regularity: key routes operate at repeatable departure windows, with two-hour connection pulses at selected hubs and clear first/last-flight policies.
- Coverage: connect regional economies directly where rail journeys are long or indirect; link Belfast and Dublin to British regional centres, and make the Isle of Man a proper network node.
- Proportionate aircraft: use 9-19 seat aircraft on thin routes and a larger 19-30 seat regional aircraft on stronger corridors. Avoid flying large jets on lightly demanded regional links.
- One passenger layer: a common timetable, through-booking where commercial agreements permit, protected connections, disruption rebooking and consistent baggage/ID information.
- Industrial opportunity: specify a British-designed high-quality commuter aircraft programme only after demand is demonstrated; treat this as a possible UK aerospace programme, not an existing BAE Systems civil-aircraft offering.

What it cannot promise

An aircraft may be airborne for 45-80 minutes, but the whole trip includes getting to the airport, security, boarding, baggage and travel from the destination airport. For many pairs, especially where a direct train is already frequent, rail will remain more convenient and have lower emissions. Fare competitiveness depends heavily on load factor, aircraft lease and maintenance cost, airport fees, fuel, crew, and the price of the rail alternative.

Initial operating scale (planning assumption)

Pilot network	Aircraft	Frequency target	Decision gate
8 routes / 8 city pairs	10-12 aircraft incl. operational spare cover	2-4 return trips per route per day, route-dependent	Expand only when demand, cost and reliability pass pre-set targets

The report uses indicative flight times and route concepts, not confirmed schedules or forecasts. CAA airport traffic data are used to understand the uneven scale of the airport system; actual route demand must be tested using market research and ticketing/booking data. [3]

1. Network architecture

The design should be a connected mesh with a light hub-and-spoke layer. A pure hub-and-spoke model forces too many passengers to double back; a fully connected mesh creates too many thin routes. The proposed compromise is direct regional trunk services where demand is credible, plus protected connections at selected interchange airports.

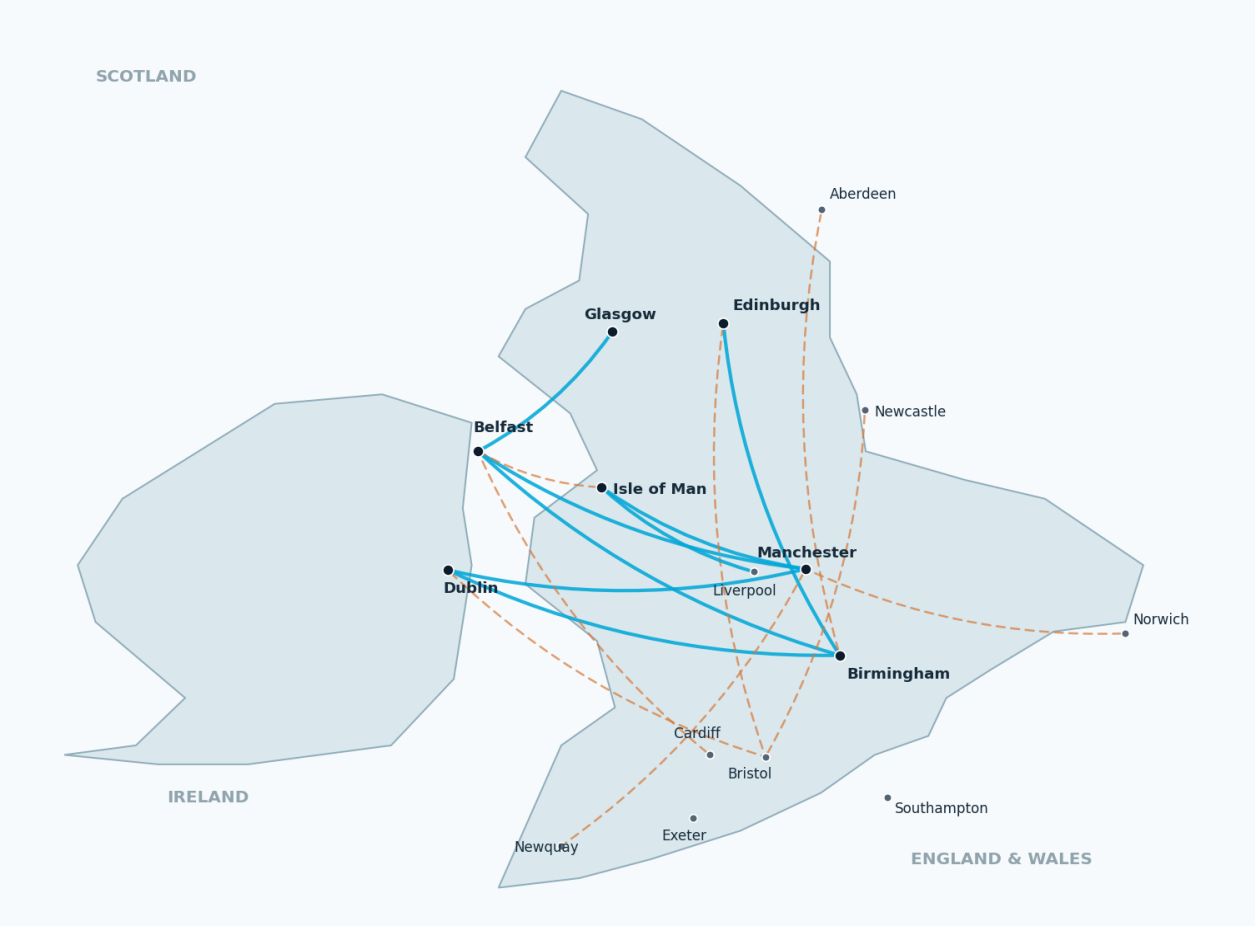


Figure 1. Illustrative network schematic. Solid cyan lines are proposed pilot corridors; dashed amber lines are expansion candidates. Line placement is schematic and must not be used for navigation or as evidence of current services.

Interchange roles

Node	Role	What it connects	Design implication
Manchester	Primary north-west interchange	Northern England, Scotland, Ireland and the Isle of Man	Prioritise easy terminal transfers and early/late operating access
Birmingham	Midlands interchange	Midlands, south-west/north-east cross-country links, Ireland	Use for journeys that otherwise route via London or require long rail journeys
Edinburgh / Glasgow	Scottish gateway pair	Central Belt, north-east Scotland and Belfast	Do not treat the two airports as interchangeable; surface transfers are required
Belfast City / Dublin	Cross-border gateways	Northern Ireland and Republic of Ireland with regional Britain	Publish clear ID, baggage and transfer information; respect separate airport/security processes
Isle of Man	Island connector	Manchester, Liverpool, Belfast and Dublin	Protect minimum connectivity and build timetable resilience for weather disruption

Do not use London as the mandatory national interchange. London airports should remain useful optional destinations, but the core purpose is to improve east-west, north-south and Irish Sea connectivity between regional economies. This is consistent with the rationale behind the DfT Union Connectivity Review's call for a more strategic national network. [2]

2. Pilot network: the first eight corridors

The first timetable should test the concept on routes with a clear geographic or island advantage, rather than opening a large number of routes at once. Frequencies below are proposed planning targets. Existing airlines may already serve some city pairs; implementation would require analysis of current capacity, schedule overlap, fare history and local airport access.

City pair	Priority	Target frequency	Indicative block time	Reason to test
Manchester - Belfast City	Core	4 return trips/day	~50-60 min	High Irish Sea / business link; direct cross-border service is time-efficient.
Manchester - Dublin	Core	4 return trips/day	~55-65 min	Large city pair and gateway route; test business and visiting-friends demand.
Birmingham - Belfast City	Core	3 return trips/day	~55-65 min	Creates a non-London Midlands to Northern Ireland corridor.
Birmingham - Dublin	Core	3 return trips/day	~60-70 min	Midlands-Ireland market, with timed connections from other regions.
Glasgow - Belfast City	Core	3 return trips/day	~40-50 min	Short Irish Sea route; weather and surface access matter to reliability.
Manchester - Isle of Man	Core	3-4 return trips/day	~40-50 min	Island connectivity; coordinate with Liverpool and Belfast services.
Liverpool - Isle of Man	Core / test	2-3 return trips/day	~30-40 min	Short crossing; proceed only if aircraft size and airport-to-town access make sense.
Edinburgh - Birmingham	Core / test	2-3 return trips/day	~65-80 min	North-south alternative to a longer rail journey; compare full door-to-door time.

Pilot selection rules

- Market test: confirm year-round origin-destination demand, business travel, tourism, visiting-friends-and-relatives and connecting flows; seasonality should be explicit.
- Airport test: confirm terminal opening hours, runway performance, noise constraints, handling availability, maintenance recovery and public transport/taxi access.
- Operational test: schedule enough recovery time to avoid cascading delays; thin routes should not be built around the assumption that every flight will run exactly on time.
- Commercial test: compare likely net yield and seat factor with route-specific costs. Do not open a route simply because it is geographically possible.

Why eight routes? It is enough to demonstrate timed connections and test multiple market types, while still allowing the operator to concentrate aircraft, crew, maintenance and sales activity. The 10-12 aircraft planning range is preliminary and depends on the final aircraft mix, flight blocks, maintenance programme and reserve policy.

3. National expansion: routes across Great Britain

After the pilot, expand into a second group of corridors selected for journey-time advantage and regional economic relevance. Some may only justify two daily return services or seasonal operation at first. These are market hypotheses, not assertions that each route has sufficient demand today.

Candidate corridor	Indicative block time	Rationale / key test
Edinburgh - Bristol	~75-90 min	4-5h+ rail is a likely comparison band; strong only if airport access and fares work.
Bristol - Newcastle	~70-85 min	North-east / south-west connection that avoids routing via London or a long rail trip.
Aberdeen - Birmingham	~75-90 min	Connects energy, engineering and business markets; test corporate commitments.
Manchester - Newquay	~60-75 min	South-west tourism and business corridor; seasonal demand and diversion options critical.
Cardiff - Belfast City	~55-70 min	Potential Wales-Northern Ireland route; airport access and demand should be proved first.
Dublin - Bristol	~60-75 min	Ireland to south-west England; coordinate with Manchester/Birmingham banks.
Norwich - Manchester	~60-75 min	East Anglia to north-west; test against rail and road including airport access.
Edinburgh - Exeter	~90-105 min	Long-distance cross-country candidate; likely second/third-phase after demand validation.
Southampton - Newcastle	~75-90 min	Potential regional business and leisure link, subject to the local catchment.
Inverness - Manchester	~85-100 min	Could strengthen access to the Highlands; seasonal, weather and diversion risks matter.
Belfast City - Bristol	~60-75 min	Regional UK link with good network value if it does not dilute stronger markets.
Isle of Man - Dublin	~40-55 min	Regional connectivity candidate; airport pair and aircraft economics must be validated.

Route hierarchy

- Tier A - core: four to six departures per direction each day on proven routes; span the business day and support multiple onward connections.
- Tier B - regional: two to four daily departures each way; schedule in connection banks and protect early-morning / late-afternoon travel where it matters.
- Tier C - lifeline or trial: one or two daily rotations, seasonal or PSO-backed if justified; never advertise it as turn-up-and-go.

The Civil Aviation Authority's 2025 airport statistics show the scale gradient: large airports such as Manchester and Edinburgh handle tens of millions of passengers, while several smaller regional airports handle hundreds of thousands or less. Smaller airports can be valuable for targeted connections, but passenger throughput alone does not prove a new air route's viability. [3]

4. The timetable: clockface, pulses and protected transfers

The timetable is the product. A route map without useful departure times is not an integrated network. The recommended system uses route-level clockface windows, then bundles arrivals and departures at selected hubs into repeatable connection waves. Frequencies must still follow demand and aircraft availability.



Figure 2. Conceptual pulse: arrivals are grouped into a window, then connecting departures leave after a protected transfer period. The drawing illustrates the principle, not a final airport slot plan.

Proposed service standards

Service class	Typical departures each way	Timetable shape	Connection promise
A - core	4-6 daily	Roughly every 2-3 hours, with first and last flights aimed at same-day business trips	45 min domestic connection target; protected through-ticket on one booking where possible
B - regional	2-4 daily	Morning / midday / late-afternoon windows; coordinate with hub waves	45-60 min connection window; avoid impossible cross-terminal transfers
C - lifeline / trial	1-2 daily	Set times aligned to essential appointments and onward transport	Longer recovery and re-accommodation plan; clear warning of limited frequency

Timetable design rules

- Common timetable epoch: use standard local times and publish the whole network for the same season, with a single feed for websites, apps and journey planners.
- Connection banks: schedule selected hub arrivals around :20-:40, with onward departures around :20-:40 of the following hour; the exact wave is airport-specific.
- Recovery margin: build 10-15 minutes of operational resilience into rotations on top of safe minimum turn times, and carry spare aircraft/crew coverage.
- Through-ticketing: sell one itinerary for connecting flights only when airlines accept connection protection, baggage transfer, disruption rules and customer-service responsibilities.
- Surface integration: align airport coach/train arrivals, publish realistic city-centre transfer times, and show a total door-to-door estimate alongside flight time.

A single national operator is not essential. A timetable coordinator, common data standards, interline agreements and consistent minimum passenger-service rules could allow multiple airlines to participate. Competition policy and commercial confidentiality would need to be respected.

5. Illustrative operating day

This example shows the logic of a repeatable service day. Times are indicative local times and are not slot-checked. The final schedule would be built with actual seasonal wind patterns, runway availability, crew duty limits, airport hours, aircraft performance and market research.

Route	1st	2nd	3rd	4th
Manchester - Belfast City	06:30	09:30	13:00	17:00
Belfast City - Manchester	08:00	11:00	14:30	18:30
Manchester - Dublin	06:45	10:00	13:30	17:15
Dublin - Manchester	08:25	11:40	15:10	18:55
Manchester - Isle of Man	07:00	10:30	14:00	17:30
Isle of Man - Manchester	08:25	11:50	15:20	18:50
Birmingham - Belfast City	07:00	11:00	15:00	-
Belfast City - Birmingham	08:35	12:35	16:35	-
Birmingham - Dublin	07:15	11:15	15:15	-
Dublin - Birmingham	08:55	12:55	16:55	-
Glasgow - Belfast City	07:30	12:30	17:30	-
Belfast City - Glasgow	08:50	13:50	18:50	-
Edinburgh - Birmingham	07:45	13:15	18:00	-
Birmingham - Edinburgh	09:45	15:15	20:00	-

Connection example

Step	Target	Requirement
Arrive Manchester	08:10	Flight on schedule; transfer passenger directed to the next departure zone
Connection buffer	45-60 min	Allows for deplaning, terminal transfer, ID/security steps where required and boarding cutoff
Depart Manchester	09:00-09:15 bank	Connection is protected only if booked on a through-ticket and operator policy covers the delay

Important: The timetable rows are an illustrative service pattern, not a promise that one aircraft can operate all listed rotations. Aircraft and crew rosters must be solved together; the timetable should be generated from a fleet-rotation model, then checked for robustness and airport slot feasibility.

6. Aircraft strategy: British quality, staged procurement

The right aircraft is not necessarily a small jet. On short regional sectors, a modern twin-engine turboprop can provide useful speed with relatively modest capacity and short-turn potential. The initial operator should select certified aircraft that already exist, rather than wait for a new type to be designed and certified.

Fleet architecture

Role	Planning class	Target use	Procurement logic
Thin routes / test markets	9-19 seats	Isle of Man, low-volume lifeline markets, early pilot services	Certified aircraft with known maintenance and operational history; monitor cost per available seat
Core regional corridors	19-30 seats	Belfast/Dublin to major regional hubs, longer cross-country city pairs	Use only when demand supports the added capacity and the aircraft achieves acceptable unit cost
Future UK aircraft programme	Conceptual 24-30 seats	A later domestic manufacturing programme for UK regional routes and export markets	Launch only after a funded business case, certification partner, engine/systems suppliers and anchor customers exist

Potential “Britannia Regional” design brief - not an existing aircraft

- Capacity: 24-30 seats, two-pilot cockpit, accessible boarding solution and cabin designed for short, frequent trips.
- Performance targets: reliable operation across 250-800 km sectors, crosswind/wet-runway capability and practical performance at smaller regional airports. Exact speed, range and runway targets require a formal design study.
- Commercial priorities: dispatch reliability, quick but safe turnaround, low fuel burn per occupied seat, common parts, maintainability in Britain and predictable lifecycle cost.
- Passenger experience: quiet cabin, dependable Wi-Fi where weight/cost permit, transparent baggage allowance, simple boarding, real-time disruption support and fares that are not dependent on last-minute price spikes.
- Industrial model: British final assembly and systems integration with UK supplier participation could be explored. “BAE-grade” is used here as a quality aspiration; the report does not imply that BAE Systems has announced a new civilian regional aircraft.

A smaller certified aircraft reference point

For scale, Tecnam describes the P2012 Traveller as a twin-engine aircraft with a maximum 2+9 seating arrangement and a 75% cruise speed of about 320 km/h. This is a useful example of the existing small commuter class, not a proposed choice for all routes. A higher-capacity turboprop would need separate comparison of acquisition cost, operating cost, payload-range performance and availability. [8]

Development sequence

Start with purchased or leased, certified aircraft; validate real demand for 24-36 months; publish the operating data; then decide whether a new UK-designed type has enough domestic and export demand to justify development cost and certification risk. A new aircraft must never be treated as a prerequisite for testing the timetable concept.

7. Journey time and fares: where flying wins

The comparison should be made door-to-door, not by comparing scheduled flight time with scheduled rail time. The following is an indicative planning model; route-by-route rail timetables and airport access times must be checked at implementation.

Component	Typical planning allowance	What changes it
Travel to origin airport	20-35 min	Local geography, public transport, traffic and parking
Pre-flight processing	30-45 min before departure	Airport security, check-in/bag drop, ID and operator rules
Flight block time	40-100 min for proposed regional routes	Distance, aircraft, weather, airspace and taxi time
Arrival / baggage	10-20 min	Carry-on only vs hold luggage; stand/gate position
Travel from destination airport	20-35 min	Distance to city centre and public transport availability

Example: 70-minute flight block + 25-minute origin access + 35-minute pre-flight buffer + 15-minute arrival/baggage + 25-minute destination access = about 170 minutes door-to-door. A rail journey of three hours may therefore be competitive; a five-hour rail journey may not be. These estimates are illustrative, not route guarantees.

Proposed fare ladder

Product	Illustrative one-way fare	Purpose
Advance / limited allocation	£39-£59	Visible entry price; only a share of seats can be sold at this level
Standard economy	£59-£99	Main fare band on routes where costs and demand permit
Flexible / business	£109-£169	Change flexibility, priority and refundable conditions; not a blanket promise
Public-service fare	Route-specific	Any subsidy or fare cap must be transparently procured and costed

A £39 promotional fare is not the same as an average fare that makes the route viable. To offer consistently low fares, the airline must have high load factors, effective aircraft utilisation, disciplined airport charges, low disruption costs and/or an explicit public-service payment. Fares should be compared with the same-day rail price, not just the cheapest rail ticket booked weeks earlier.

8. Unit economics and the GDP case

The economic case should be assessed at three levels: (1) route-level airline economics, (2) passenger and business benefits from time saved and better connections, and (3) wider UK industrial and productivity effects. These are different measures and should not be added together without adjusting for overlap.

Illustrative route-level break-even

For a transparent first-pass calculation, assume a 30-seat aircraft, a 1.1-hour scheduled block sector, and a broad £1,500-£2,500 cost per block hour for aircraft, crew, fuel and maintenance. This is a modelling range, not a supplier quote; airport fees, central overhead, sales/distribution, financing and disruption costs can raise the required revenue.

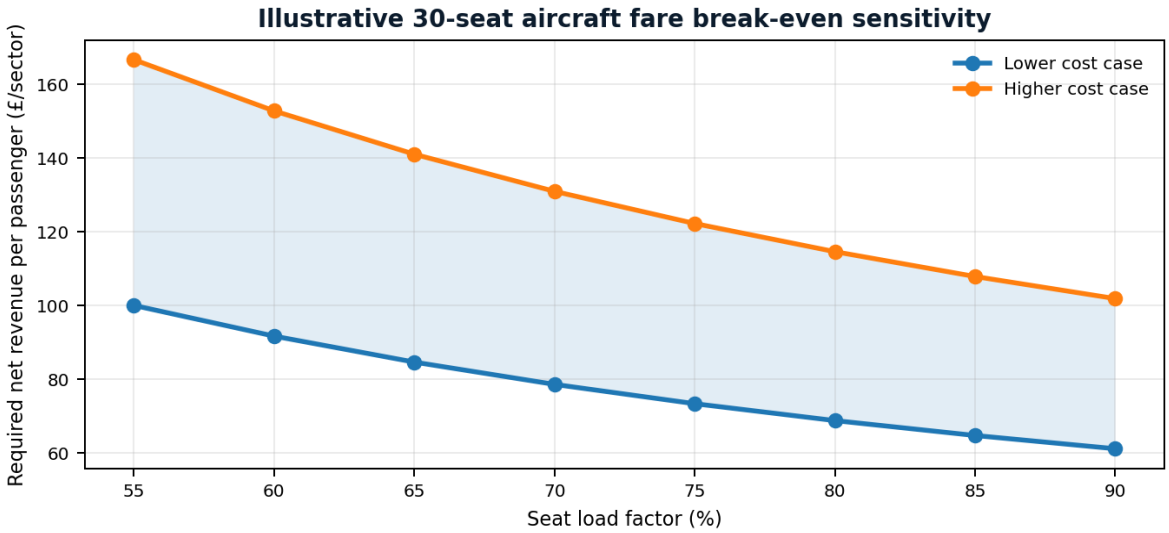


Figure 3. Net revenue required per passenger if one sector costs £1,650-£2,750 and the aircraft has 30 seats. Higher load factors reduce break-even fares. The model excludes some overheads and should be stress-tested with actual aircraft and airport quotes.

Seat load factor	Passengers on 30 seats	£1,650 sector cost	£2,750 sector cost
60%	18	£91.67/pax	£152.78/pax
75%	22.5	£73.33/pax	£122.22/pax
85%	25.5	£64.71/pax	£107.84/pax

Implication: the network may deliver fares below rail on selected markets, but it cannot assume that every route can do so. Public support should be assessed only where there is a demonstrable public-interest case and no viable commercial alternative. The DfT has an established Public Service Obligation (PSO) process; route proposals need to meet policy criteria rather than treating subsidy as automatic. [4]

GDP and productivity channels

- Business time: more same-day meetings across regions, less time spent on long cross-country trips and a larger practical catchment for specialist services.
- Labour and skills: improved access to regional labour markets and training, if schedules match actual work patterns and are affordable to residents.
- Tourism: new visitor flows to Wales, Scotland, the South West and the Isle of Man, while monitoring displacement from existing rail/air services.
- Industrial supply chain: aircraft manufacturing, maintenance, simulation, avionics, software, airport engineering, training and sustainable aviation fuel activity.

A defensible appraisal should quantify passenger time savings, reliability, induced trips, business productivity and net additional tourism spend. The existing £14 billion aviation contribution to GDP cited by the DfT is a sector baseline, not a multiplier to apply to this proposal. [1]

9. Climate, regulation and cross-border operation

Regional air services create a real trade-off: they can save time on long, indirect journeys, but a turboprop flight will generally have higher emissions per passenger than a well-used train on many domestic corridors. The public business case must include emissions and surface access, not only journey time and jobs.

Climate and environmental design requirements

- Route selection: prioritise journeys with a substantial time advantage, poor direct rail alternatives, island geography or clear connectivity gaps; avoid subsidising short hops that duplicate strong rail services without a measurable benefit.
- Occupancy: do not operate oversized aircraft on thin routes. Report load factor and emissions per passenger-kilometre, with empty positioning legs included.
- Fuel transition: assess sustainable aviation fuel availability and lifecycle emissions; avoid labelling SAF as zero-emission. Monitor future hybrid-electric and hydrogen aircraft only as technologies become certified and operationally credible.
- Airport impact: assess noise, local air quality, ground transport, night operation, and the effect of additional movements at constrained airports.
- Public reporting: publish annual route performance across punctuality, cancellations, load factor, passenger satisfaction, cost per passenger and greenhouse-gas intensity.

The UK government's Jet Zero strategy sets out a pathway for aviation decarbonisation and identifies efficiency, sustainable aviation fuels and zero-emission flight as key strands. The government also publishes annual greenhouse-gas conversion factors for reporting. A full appraisal should use the relevant 2026 factors and include the aircraft, load-factor and trip-distance assumptions. [5][6]

Regulatory path

Workstream	Requirement / issue	Practical implication
Operator licensing	CAA Air Operator Certificate (AOC) and relevant operating licence for commercial transport	CAA advises a pre-application process; planning a new operator commonly requires 6-12 months, depending on readiness and scope. [7]
Aircraft approvals	Aircraft type approval, continuing airworthiness, maintenance and crew training	Different from the airline AOC. A new aircraft programme adds a far longer and higher-risk certification path.
Route and airport permissions	Slots where required, airport hours, noise rules, handling contracts, airspace and performance restrictions	A route that works on paper may not fit airport capacity or opening-hour limits.
Ireland / Northern Ireland	International route approvals, passenger ID rules, security and baggage transfer processes	Design time buffers per airport and route; do not assume identical processes across all airports.
Passenger protection	Through-ticket terms, refunds, delay handling, accessibility and clear disruption support	A connecting network requires clear accountable ownership of the whole passenger journey.

Sources: CAA AOC guidance and application process; DfT Jet Zero; DESNZ greenhouse-gas factors. [5][6][7]

10. Implementation programme

Deliver the timetable as a staged programme with gates. The critical path is usually not the map: it is route demand, aircraft access, operating permissions, airport contracts and the ability to maintain reliable service at an acceptable average fare.

Phase	Timing (indicative)	Work	Gate to proceed
0 - evidence	0-6 months	Origin-destination demand study; airport access survey; rail/road comparator; airline and lessor market sounding; community/business consultation.	At least 8 candidate routes show credible demand and measured door-to-door advantage.
1 - commercial design	6-12 months	Select operator/consortium; negotiate aircraft, crew, maintenance, airport handling, timetable data and passenger-service agreements.	Route model passes base/downside cases; regulatory plan agreed with CAA.
2 - pilot launch	12-24 months	Launch 4-8 corridors with a limited common fleet; market fares; integrate journey planner and disruption support; measure outcomes monthly.	Load factor, unit cost, completion factor and customer satisfaction meet agreed thresholds.
3 - expansion	24-36 months	Add selected regional links, improve hub pulses and through-ticketing; review PSO routes where the public case is strong.	Route-level financial and social case remains positive with realistic fares and emissions.
4 - aircraft industry decision	36+ months	Complete feasibility study for a UK-built 24-30 seat aircraft; secure anchor orders, suppliers, certification strategy, manufacturing and service model.	Credible export demand and lifecycle economics; no dependence on unproven aircraft performance.

Proposed national timetable office

- Own the public-facing master timetable and service data standard; it need not operate the flights.
- Coordinate transfer windows and publish minimum connection standards with participating airlines and airports.
- Maintain a public dashboard for reliability, cancellations, average fares, emissions and passenger feedback.
- Run an annual route review: expand routes that work, redesign those with fixable timetable problems, and withdraw routes that remain structurally uneconomic unless a transparent public-service case applies.

Institutionally, this could be a public-private partnership involving the Department for Transport, devolved administrations, airports, airlines, rail/coach providers and business groups. Public funding should support defined public outcomes, not unconditioned route risk.

11. Principal risks and mitigations

Risk	Why it matters	Mitigation
Insufficient demand	Small aircraft have limited capacity; low seat factor quickly increases cost per passenger.	Test paid demand first; use advance bookings and business anchor agreements; launch fewer routes.
Promotional fare trap	Low introductory fares can hide a weak underlying yield.	Publish average net yield and route-level contribution; set clear fare floors and limits on discounted inventory.
Network fragility	A delay can break onward connections and strand passengers when frequencies are low.	Build recovery time; protect transfers only with a credible rebooking plan; keep spare aircraft/crew cover.
Rail competition	On strong rail corridors, flying can be slower door-to-door and more carbon-intensive.	Target long/indirect routes, islands and connectivity gaps; require a quantified journey advantage.
Weather / diversion	Irish Sea and Scottish operations can face weather-related disruption.	Use certified instrument operations, diversion planning, passenger care plans and route-specific winter schedules.
Aircraft programme cost	A new aircraft design can take years and require substantial capital before revenues exist.	Operate existing certified aircraft first; pursue the new type only with anchor buyers and a staged business case.
Airport constraints	Opening hours, local noise, runway restrictions and slots can undermine clockface promises.	Negotiate airport operating windows before publishing a route; preserve schedule resilience and consult communities.
Cross-border complexity	Security, ID, baggage and through-ticket rules vary by route and airport.	Route-level operational review, clear passenger terms and one accountable customer-service process.
Environmental objection	New flights may shift passengers from rail and increase emissions.	Publish comparative emissions and net modal-shift analysis; set route-specific environmental tests.

Recommended success measures

- Network: on-time performance, completion factor, missed connections, route frequency and same-day return availability.
- Commercial: load factor, net revenue per passenger, cost per block hour, route contribution, distribution cost and refunds/rebooking burden.
- Public value: door-to-door time saved, regional business usage, net additional visitor spend, accessibility and user satisfaction.
- Environment: emissions per passenger-kilometre, load factors, empty positioning flights and modal shift from rail/road.

12. Conclusions and recommended decisions

A national regional-air timetable is technically possible, but it must be built as an operating system, not simply a route map. Its value comes from dependable frequencies, sensible connection windows, honest door-to-door time information, useful airport access and a route portfolio shaped by evidence.

Recommended decisions

01 Approve a six-month feasibility stage for eight pilot corridors covering Manchester, Birmingham, Scotland, Belfast, Dublin and the Isle of Man.

02 Commission a route demand and journey-time study using CAA route/passenger data, actual rail timetables, airport access, fares, business travel research and local stakeholder input.

03 Specify a clockface timetable standard with two-hour connection pulses at chosen hubs, protected through-tickets where commercially supported and a common live information feed.

04 Source certified aircraft first across 9-19 seat and 19-30 seat categories; size each route for credible demand, not branding.

05 Build an open-book financial model showing base and downside assumptions for load factor, average net fare, fuel, maintenance, crew, lease/ownership, airport fees and disruption.

06 Use public-service support selectively under existing policy where a public-interest case is clear; publish the subsidy per passenger and service obligations.

07 Defer any UK-designed aircraft commitment until proven airline demand and credible anchor customers justify design, certification and manufacturing costs.

The strategic proposition
Connect Britain by air where geography and time justify it; make routes predictable enough to feel like public transport; and use real operating data to determine where a British-built regional aircraft industry could be commercially sustainable.

Final judgement

The best first version is not a nationwide airline with dozens of daily thin routes. It is a small, reliable network of regional trunk routes, coordinated connection waves and well-chosen island/cross-border services. If that pilot can show strong load factors, good reliability, fare competitiveness and a net public benefit, the network can expand across Great Britain. A new UK aircraft programme should be the result of proven demand, not the assumption on which the plan depends.

Sources, evidence and modelling notes

The following primary sources underpin the policy, traffic, regulatory and environmental context. Route choices, timetable windows, aircraft concept and economics in this report are proposed planning assumptions rather than findings from a completed feasibility study.

[1] Department for Transport (30 Jan 2025). Airport capacity and expansion: a government update. Includes the reported £14bn direct GDP contribution from air transport in 2023, jobs and the role of connectivity.
https://www.gov.uk/government/publications/airport-capacity-and-expansion-a-government-update

[2] Department for Transport (26 Nov 2021; updated Dec 2023). Union connectivity review: final report. Recommendations for a more integrated multi-modal network and regional connectivity.
https://www.gov.uk/government/publications/union-connectivity-review-final-report

[3] UK Civil Aviation Authority. Annual airport data 2025. Terminal passenger, movements and route statistics; 2025 data reported complete in March 2026.
https://www.caa.co.uk/data-and-analysis/uk-aviation-market/airports/uk-airport-data/uk-airport-data-2025/annual-2025/

[4] Department for Transport. Public service obligation: regional air connectivity protection. Policy and criteria for domestic air-service PSOs.
https://www.gov.uk/government/publications/public-service-obligation-regional-air-access-to-london

[5] Department for Transport (19 Jul 2022). Jet Zero strategy: delivering net zero aviation by 2050.
https://www.gov.uk/government/publications/jet-zero-strategy-delivering-net-zero-aviation-by-2050

[6] Department for Energy Security and Net Zero (11 Jun 2026; updated 31 Jul 2026). Greenhouse gas reporting: conversion factors 2026.
https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2026

[7] UK Civil Aviation Authority. Air operator certificates and Preparing and applying for an Air Operator Certificate.
https://www.caa.co.uk/commercial-industry/aircraft/operations/air-operator-certificates/air-operator-certificates/

[8] Tecnam. P2012 Traveller specifications. Example of a twin-engine small commuter aircraft: maximum 2+9 seats, 320 km/h 75% cruise specification. Used as a class reference only.
https://tecnam.com/aircraft/p2012-traveller/

[9] Department for Transport (18 Jun 2026). UK aviation forecast 2026. Long-term passenger, movements and emissions forecast context.
https://www.gov.uk/government/publications/uk-aviation-forecast-2026

Modelling notes

- All proposed route block times are rounded planning estimates for schedule design. They are not live timetable data and should be validated with operator flight planning.
- All proposed frequencies and flight times require airport/slot feasibility, aircraft rotation planning, current route demand, market testing and a seasonal schedule study.
- Fares and hourly operating costs are scenario inputs selected to make the economics transparent. They are not quotations or forecasts. The cost model omits some route-specific overheads and needs replacement with actual aircraft, airport, handling, fuel and labour quotes.
- The map and timetable illustrations are conceptual. They are not navigational charts or actual schedules.
- GDP effects are not quantified in this concept report. A full appraisal must distinguish net additional productivity and consumer benefits from existing activity, transfers from other modes and displacement between regions.